

BOUDROUSS RÉDA | CV

Software Engineer – OCaml & Rust, WebAssembly, Static Analysis – Paris

github.com/rboudrouss
linkedin.com/in/rboudrouss
contact@rboud.com
rboud.com
+33 (0)7 55 78 85 16

Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, specialized in **static analysis**, **abstract interpretation** and the implementation of **programming languages** and their runtimes. Much of my personal work sits at the **OCaml/WebAssembly** boundary, between the MOPSA analyzer (OCaml/C/C++) I cross-compiled to Wasm and the analyzer I wrote in **Rust**. At Hexaglobe I ship production **GitLab CI/CD** and test infrastructure for video-streaming systems.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python written in **OCaml/C/C++**, to **WebAssembly** by compiling the OCaml runtime and all native dependencies (Clang frontend, Apron, GMP, MPFR) with Emscripten and running MOPSA's OCaml bytecode on top.
- Worked on both sides of the **OCaml runtime / C FFI** boundary, generating a static table of ~1,400 C primitives (no dlopen on Wasm) and fixing an unsigned-wraparound bug in the runtime's Tag_val macro. Recompiled Apron's numerical domains to use exact GMP rationals, addressing the **soundness** hazard of Wasm's fixed rounding mode.
- Shipped a fully client-side **playground** (Web Workers, SharedArrayBuffer + Atomics for an interactive REPL) and a technical **write-up**.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Designed and implemented an **abstract-interpretation**-based static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.
- Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a dedicated CFG-based IR and flags an infinite render loop when the state fixpoint **widens**. Resolves and inlines custom hooks across files, with concrete semantics based on the React-tRace paper (OOPSLA 2025).

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: **static analysis by abstract interpretation** and type systems (course led by Antoine Miné), program specification & formal verification, design & implementation of programming languages (interpreters vs. compilers), **concurrent & distributed programming**.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases and functional programming.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming)

2025 – 2026

- Built complete **GitLab CI/CD** pipelines and shared CI components (npm publishing, centralized Allure reporting) consumed by 4+ repositories.
- Designed **test strategies** (unit, integration, E2E) for zero-coverage legacy products, reaching ~**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS).
- Industrialized **AI-agent test production** with a verification-first approach built on sandboxed multi-agent orchestration, versioned guardrails, mandatory green tests and systematic human review.
- Built objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking) and built a **local-first** mobile web app. As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.